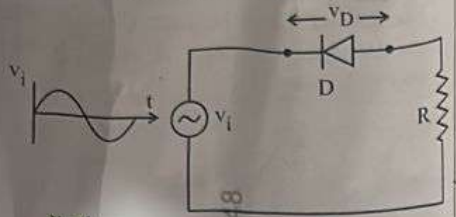
The speed of light in vacuum is taken as unity. If light takes 6 min 40 s to reach the Earth from the Sun, the distance between the Sun and the Earth in new unit is :

(1) 500 (2) 3×10^8

(3) 400 (4) 3×10^{10}

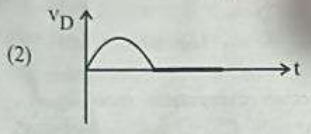
In the circuit shown below, the voltage appearing across the diode D will be of the form :



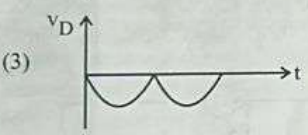
(1)



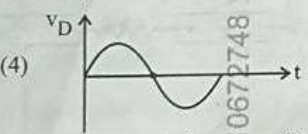
(2)



(3)



(4)



10. A submarine is designed to withstand an absolute pressure of 100 atm. How deep can it go below the water surface ?
(Consider the density of water = 1000 kg m^{-3}, 1 atm = 1×10^5 Pa and gravitational acceleration g = 10 m/s^2)

(1) 990 m (2) 9000 m

(3) 99 m (4) 9900 m

11. An electric heater supplies heat to a system at a rate of 100 W. If the system performs work at a rate of 75 J/s, then the rate at which internal energy increases will be :

(1) 125 W (2) 75 W

(3) 100 W (4) 25 W

12. A 100-turn closely wound circular coil of radius 5 cm has a magnetic field of 3.14×10^{-3} T at its centre. The current flowing through the coil, and the magnitude of the magnetic moment of this coil are, respectively :
(Take $\mu_0 = 4\pi \times 10^{-7}$ T m/A)

(1) 2 A, 4 A m^2 (2) 2·5 A, 20 A m^2

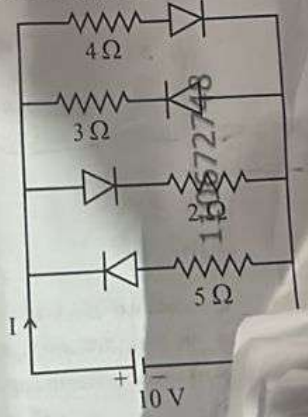
(3) 2·5 A, 2 A m^2 (4) 2 A, 10 A m^2

13. In Young's double slit experiment, using monochromatic light of wavelength λ, the intensity of light at a point on the screen where the path difference is λ, is K units. The intensity of light at a point where the path difference is $\frac{\lambda}{3}$ will be :

(1) $\frac{K}{2}$ (2) 2 K

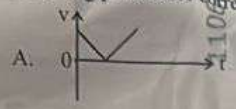
(3) $\frac{K}{4}$ (4) K

14. The current I in the circuit shown below is :
(All diodes are ideal and identical)

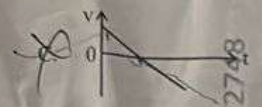


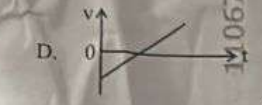
(1) $\frac{5}{3}$ A (2) $\frac{5}{9}$ A

1. The following plots show variation of velocity (v) with time (t), of a ball thrown vertically upward, and falling back. Which of the following plots is/are correct ?

A. 

B. 

C. 

D. 

E. 

(1) B only (2) A and E only
(3) D only (4) C only

2. For a metal of work function 6·6 eV, which of the following wavelengths of incident radiation does *not* give rise to the photoelectric effect ?
(Take Planck's constant as $6·6 \times 10^{-34}$ J s)
(1) 50 nm (2) 100 nm
(3) 150 nm (4) 200 nm

3. The power of a crane, which lifts a mass of 1000 kg to a height of 20 m in 10 s is :
($g = 9·8$ m/s^2)
(1) 39·2 kW (2) 39·2 W
(3) 19·6 kW (4) 19·6 W

4. Match List I with List II :

List I	List II
A. $E = h\nu$	I. de Broglie wavelength
B. Diffraction and Interference	II. Particle nature of light
C. $\lambda = h/p$	III. Wave nature of light
D. Compton effect	IV. Energy of photon

Choose the **correct** answer from the options given below :
(1) A-IV, B-I, C-II, D-III
(2) A-I, B-IV, C-III, D-II
(3) A-IV, B-III, C-II, D-I
(4) A-IV, B-III, C-I, D-II

5. The magnitude and direction of the acceleration produced in a body of mass 5 kg when two mutually perpendicular forces 8 N and 6 N act on it, are respectively :
(1) 2 m s^{-2}; $\tan^{-1}(4/3)$ with 8 N force
(2) 2 m s^{-2}; $\tan^{-1}(3/4)$ with 8 N force
(3) 2 m s^{-2}; $\tan^{-1}(3/4)$ with 6 N force
(4) 20 m s^{-2}; $\tan^{-1}(4/3)$ with 8 N force

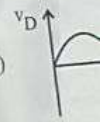
6. The sum of kinetic energy and potential energy of a simple pendulum bob is 0·02 joule. The speed of the simple pendulum bob at equilibrium position is approximately :
(Consider mass of the bob = 20 g)
(1) 14·1 m/s (2) 1·41 m/s
(3) 2·0 m/s (4) 0·2 m/s

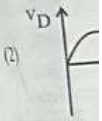
7. A box of mass 15 kg is kept on the floor of a stationary trolley. The coefficient of static friction between the box and the trolley is 0·12. Keeping the box in stationary state over the trolley, the maximum acceleration with which the trolley can be moved horizontally in m s^{-2} is :
($g = 10$ m/s^2)
(1) 1·2 (2) 1·8
(3) 1·5 (4) 2·1

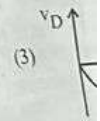
The speed of light in
ight takes 6 min 40
Sun, the distance bet
ew unit is :
(1) 500
(3) 400

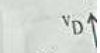
In the circuit show
across the diode D



(1) 

(2) 

(3) 

(4) 

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(1)
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15. In a concave lens, a ray of light emanating from the object parallel to the principal axis of the lens, after refraction :

(1) passes through the second principal focus.

(2) appears to diverge from the first principal focus.

(3) emerges parallel to the principal axis.

(4) passes through 2F, which is the radius of curvature of the lens.

16. A galvanometer of resistance 100 Ω gives full scale deflection for a current of 1 mA. It is converted into an ammeter of range 0 – 10 A. The shunt required is :

(1) 0·10 Ω (2) 0·001 Ω

(3) 1·0 Ω (4) 0·01 Ω

17. In the first excited state of hydrogen atom, the energy of its electron is – 3·4 eV. The radial distance of the electron from the hydrogen nucleus in this case is approximately :

(Take 1 eV = $1{\cdot}6 \times 10^{-19}$ J, e = $1{\cdot}6 \times 10^{-19}$ C and $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ N\,m^2/C^2$)

(1) $2{\cdot}1 \times 10^{-11}$ m (2) $2{\cdot}1 \times 10^{-10}$ m

(3) $2{\cdot}1 \times 10^{-9}$ m (4) $2{\cdot}1 \times 10^{-8}$ m

18. The amount of work done to raise a mass 'm' from the surface of the Earth to a height equal to the radius of the Earth 'R', will be :

(1) mg R (2) 2 mg R

(3) $mg\frac{R}{4}$ (4) $mg\frac{R}{2}$

19. An ac circuit contains a resistance of 1 kΩ, a capacitor of 0·1 μF and an inductor of 1 mH connected in series. The resonance frequency of the circuit is approximately :

(1) 13·5 kHz (2) 15·9 kHz

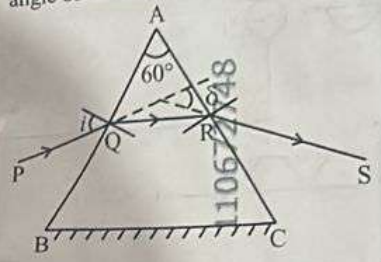
(3) 10·1 kHz (4) 20·7 kHz

20. Consider two uncharged capacitors of same capacitance 200 pF. One of them is charged by a 100 V supply and disconnected. Now the capacitor is connected to the uncharged capacitor. The amount of electrostatic energy lost in process is :

(1) 1·0 J (2) 0·5 J

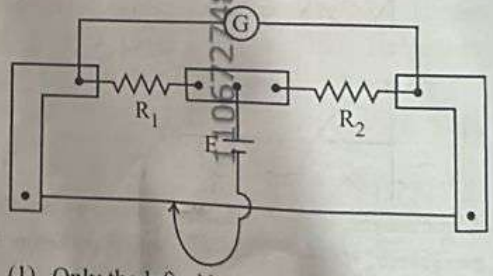
(3) $1{\cdot}0 \times 10^{-6}$ J (4) $0{\cdot}5 \times 10^{-6}$ J

21. A ray of monochromatic light is passing through an equilateral prism (ABC) as shown in the figure. The refracted ray (QR) is parallel to its base (BC) and the angle of incidence (i) is 50°. Then the angle of deviation (δ) is :



(1) 45° (2) 40° (3) 35° (4) 55°

22. In a metre bridge experiment (see figure), the positions of the cell, E, and galvanometer, G, are interchanged. We shall observe in the galvanometer :



(1) Only the left-sided deflection

(2) Both right-sided and left-sided deflection and at balance point, no deflection

(3) Only the right-sided deflection

(4) There will be no deflection irrespective of the position of the jockey